

CSE471 — DEEP LEARNING FOR COMPUTER VISION | WEEK 6 | LECTURE 11 | UNIT III

Transfer Learning in Computer Vision

Why teach a model everything from scratch?

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If a neural network has already learned useful visual knowledge, why throw that knowledge away?

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Learning Objectives

- Explain why training a deep CNN completely from scratch can be difficult.
- Explain the basic idea of transfer learning in simple words.
- Define a pretrained model.
- Explain what a pretrained backbone is.
- Understand why knowledge learned from one dataset transfers to another.
- Describe the basic idea behind ResNet.
- Describe the basic idea behind EfficientNet.
- Describe the basic idea behind MobileNet.
- Compare ResNet, EfficientNet, and MobileNet conceptually.
- Identify situations where transfer learning is highly useful.

Quick Classroom Question

Imagine teaching a child to recognize dogs. Do you first teach them:

Starting from Zero?

- What an edge is?
- What a shape is?
- What texture looks like?

Using Prior Knowledge?

Or do they already know these fundamental visual building blocks from their previous life experiences?

Section 1: Why Not Train From Scratch?

Understanding the limitations of zero-knowledge initialization.

What Does "Training From Scratch" Mean?

Training a model from scratch means starting with **randomly initialized weights** and learning the required patterns entirely from the new dataset.



1. Random Weights

The network is completely "blank" and guesses randomly.



2. Target Data

It slowly adjusts math based purely on the provided dataset.

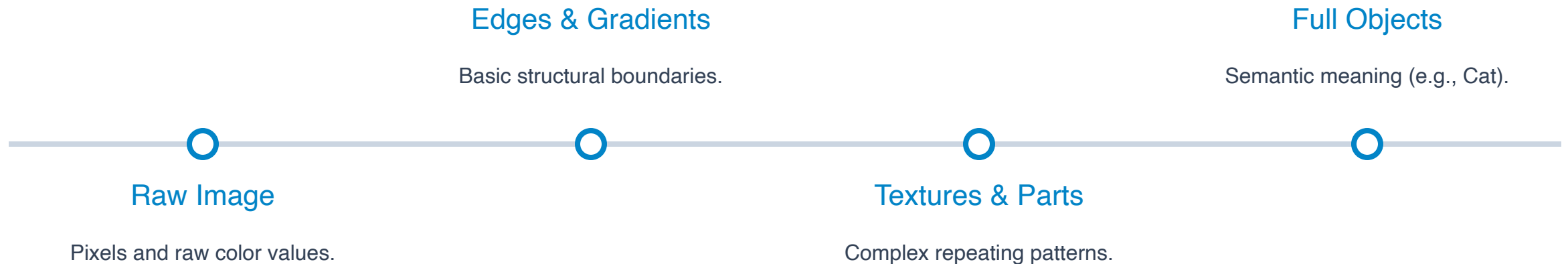


3. Final Model

It eventually learns features directly from the ground up.

What Happens Inside a CNN?

A conceptual hierarchy of visual learning. Neural networks learn concepts sequentially, from basic math to semantic meaning.



Instructor Note: Misconception alert! Clarify that this is highly conceptual; the math doesn't guarantee a literal 'ear' detector, but it statistically builds complexity layer by layer.

Time: 3 mins

The Problem With Starting From Zero

-  **Massive Data Requirement:** Needs millions of labeled images to reliably learn basic edges up to complex objects without memorizing.
-  **Time Consuming:** Can take days or weeks of continuous processing to train from completely random weights.
-  **Computationally Expensive:** Requires highly powerful, expensive GPUs (e.g., NVIDIA A100 clusters).
-  **Prone to Overfitting:** On small datasets, the model memorizes the training data rather than learning general, useful visual rules.

Real-World: Medical Imaging

Goal: Identify a rare disease from X-ray images.

Problem: Only a few thousand labeled images exist globally. If we train from random weights, the CNN fails to learn deep visual patterns.

Can we reuse what a general model already knows about shapes and textures?



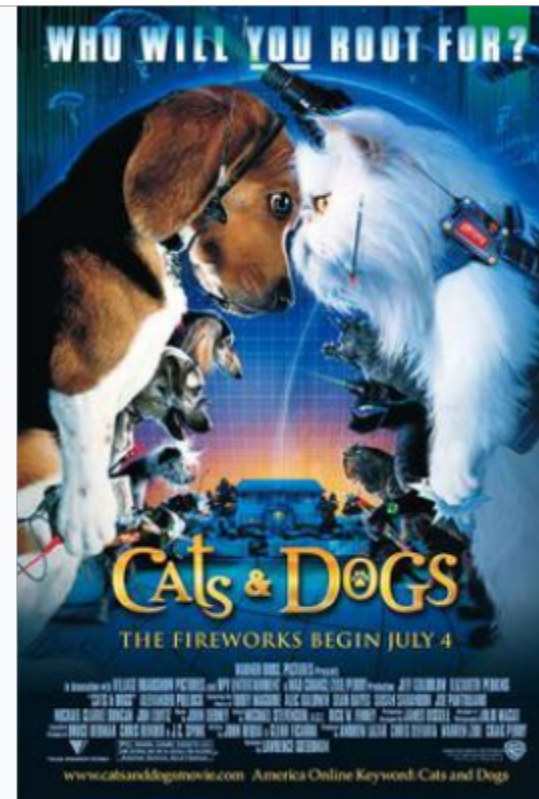
Real-World: Cats vs. Dogs

Imagine two datasets:

- **Dataset A:** Millions of general internet images (cars, trees, animals, buildings).
- **Dataset B:** A small dataset of only cats and dogs.

A CNN trained on Dataset A has already spent weeks learning how to see edges, fur textures, and animal shapes.

By transferring this CNN to Dataset B, it only needs to learn the *difference* between a cat and a dog, not how to see.



What If We Only Have Limited Data?

1,000

Labeled Images Available

The Industrial Scenario

You work for a manufacturing company that has 1,000 labeled images of defective products.

Question: Would you prefer to train a large CNN from random weights?

Answer: Usually, we can do much better by starting from a Pretrained Model.

Section 2: Introducing Transfer Learning

Reusing computational knowledge for faster, better results.

The Core Idea



Transfer learning is the process of using knowledge learned from one task or dataset to help solve another related task.



Learn Once → Reuse Useful Knowledge → Adapt to New Problem

The Human Analogy

Human Learning

A student already knows English syntax, basic logic, and algebra.

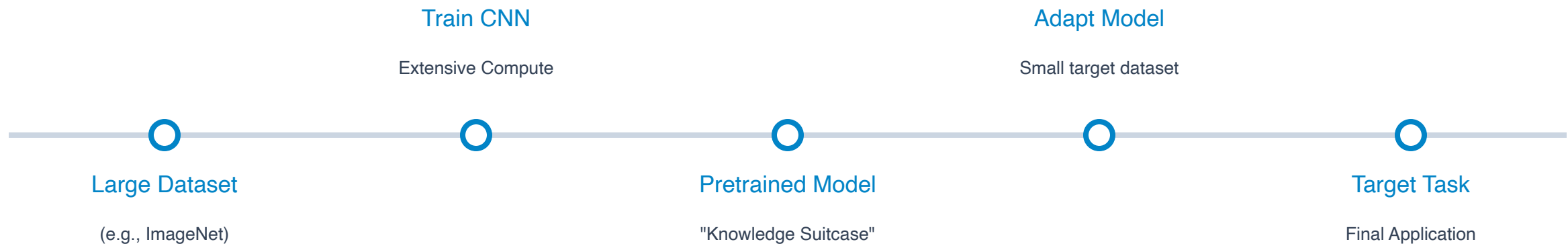
Because of this prior knowledge, they can learn Python programming **very quickly** compared to a toddler starting from zero.

Machine Learning

A CNN already knows edges, geometric textures, and shapes from millions of general images.

Because of this prior knowledge, it learns to classify a new medical image **very quickly** compared to random weights.

Transfer Learning Pipeline



Instructor Note: Walk through the pipeline step-by-step. Point out that the first two steps are usually done by massive tech companies, and we (the engineers) take over at step 3.

Time: 3 mins

What is a Pretrained Model?

-  Definition: A model that has already been trained on a large dataset for a previous task.
-  Saved Parameters: It has already learned useful weights (parameters). We don't need to initialize with random numbers.
-  Knowledge Suitcase: Think of it as a suitcase of visual tools the model brings to its new job.
-  Important Caveat: The model does NOT "know" objects the way humans do. It simply encodes useful statistical pixel patterns.

Section 3: Pretrained Backbones

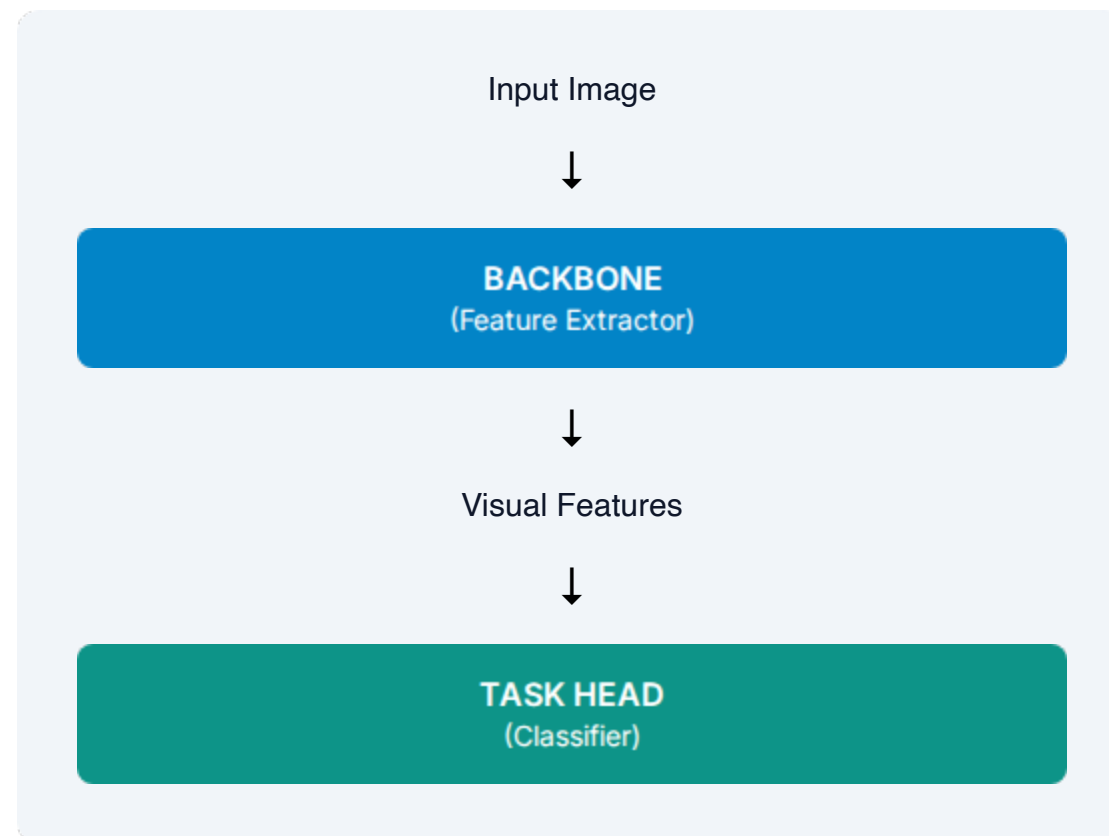
The visual engines powering modern AI applications.

What is a Backbone?

A **backbone** is the main feature-extracting part of a neural network.

It acts as the visual engine of the system. The backbone does the heavy lifting of understanding the image geometry, textures, and structures.

The **Head** is a smaller, final network layer adapted specifically to your output classes (e.g., Cat vs Dog).



Why Use a Pretrained Backbone?

- ↓ Requires significantly less data: The hard work of learning visual rules is already done.
- 🕒 Reduces training time: Drops from days/weeks down to minutes or hours.
- 🔧 Lowers compute requirements: Cheaper hardware can be used to just train the "head".
- 🎯 Provides a robust starting point: Mathematically avoids getting "stuck" in poor local minima during training.

Section 4: Meet ResNet

The architecture that made deep networks possible.

The Degradation Problem

ResNet (Residual Networks) was introduced to solve a specific issue.

Before ResNet, researchers noticed that simply adding more layers to a deep network made training harder, and accuracy actually decreased.

Information and gradients got "lost" as they passed through too many sequential layers.



The Solution: Skip Connections

ResNet in one sentence: It makes very deep networks easier to train by allowing information to skip some layers through shortcut connections.



Without Skip

Every piece of data is forced to transform through every single narrow layer.



With Skip

Information can take a "highway shortcut" forward, preserving original data.



Variants

ResNet-18, ResNet-34, ResNet-50. The number indicates the depth of the network.

Section 5: EfficientNet

Balancing accuracy and computational scale.

The Big Idea: Compound Scaling

EfficientNet focuses on achieving strong performance while using computational resources efficiently.

Instead of randomly making a network deeper or wider, EfficientNet scales three things together systematically:

- **Depth:** Number of layers.
- **Width:** Number of channels.
- **Resolution:** Image size.

The EfficientNet Family

B0: Smallest, fastest



B3: Balanced mid-tier



B7: Largest, most accurate, highly compute-intensive

Section 6: MobileNet

Deep learning for the edge.

Lightweight Models for Edge AI

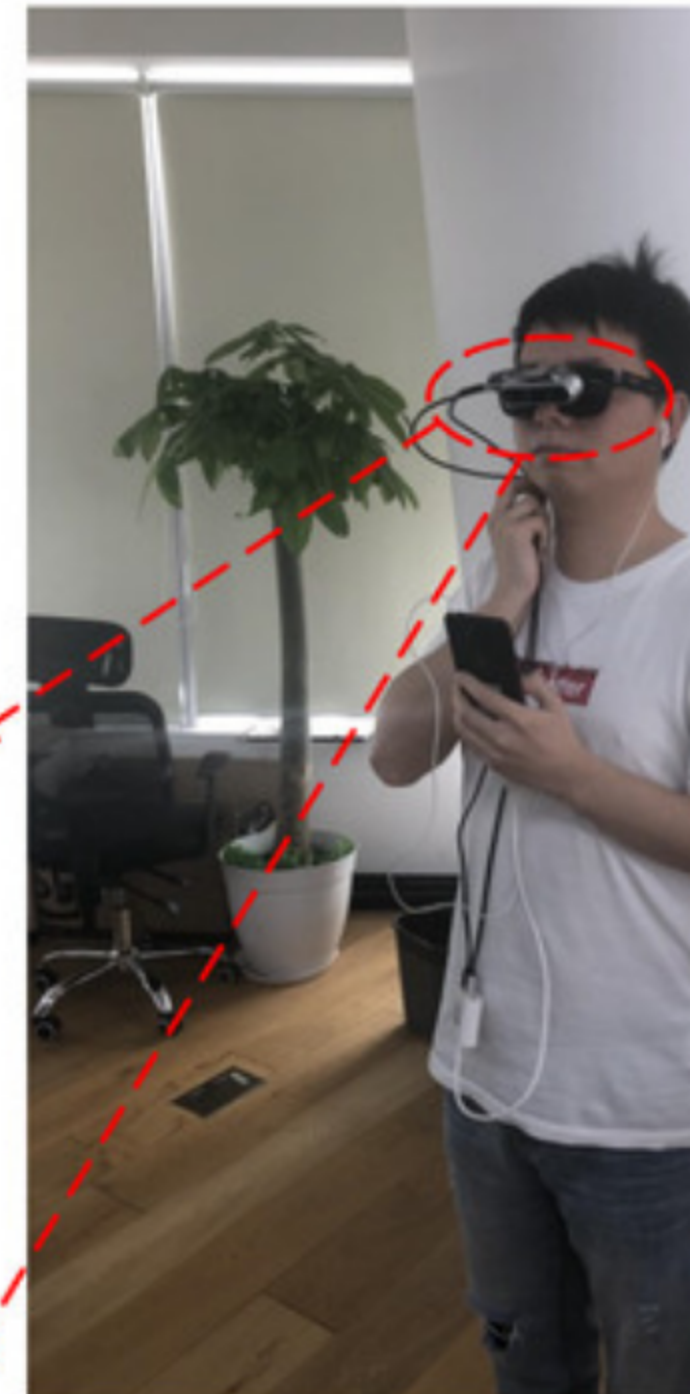
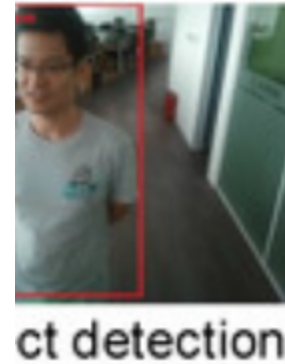
MobileNet is designed specifically for efficient deep learning on devices with limited computational resources.

Target Hardware:

- Smartphones & Tablets
- Edge AI devices (IoT)
- Embedded systems & robotics

It uses *Depthwise Separable Convolutions* to split heavy mathematical operations into two smaller, highly efficient steps.

Instructor Note: Ask students: "Does the most accurate model always make the best product visually."



Comparing the Architectures

Feature	ResNet	EfficientNet	MobileNet
Main Idea	Skip connections	Compound scaling	Depthwise separable conv.
Main Strength	Reliable, easy to train	Accuracy/Compute balance	Extremely lightweight
Typical Use	General purpose baseline	High accuracy constraints	Phones, IoT, Robotics
Compute Cost	Moderate to High	Optimized / Scalable	Very Low

Instructor Note: Walk through the table. This is an excellent summary for their notes. Help them build an engineering mindset: engineering is about trade-offs. **Time: 3 mins**

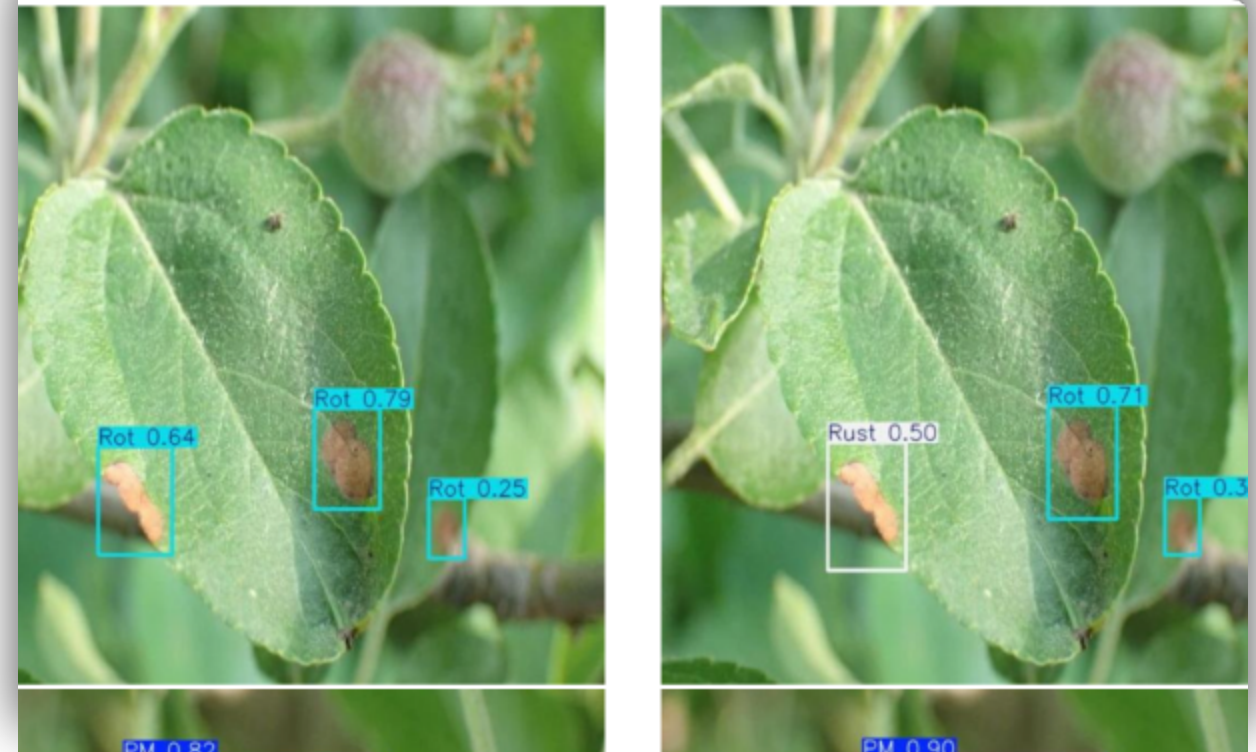
Case Study: Plant Disease

Problem: Build an AI system that identifies plant diseases from leaf images for farmers.

Constraint: Small dataset of only 2,000 images.

Solution:

1. Choose a pretrained backbone (e.g., ResNet-50).
2. Extract useful visual features from the leaves.
3. Train a new classification head adapted to disease classes.



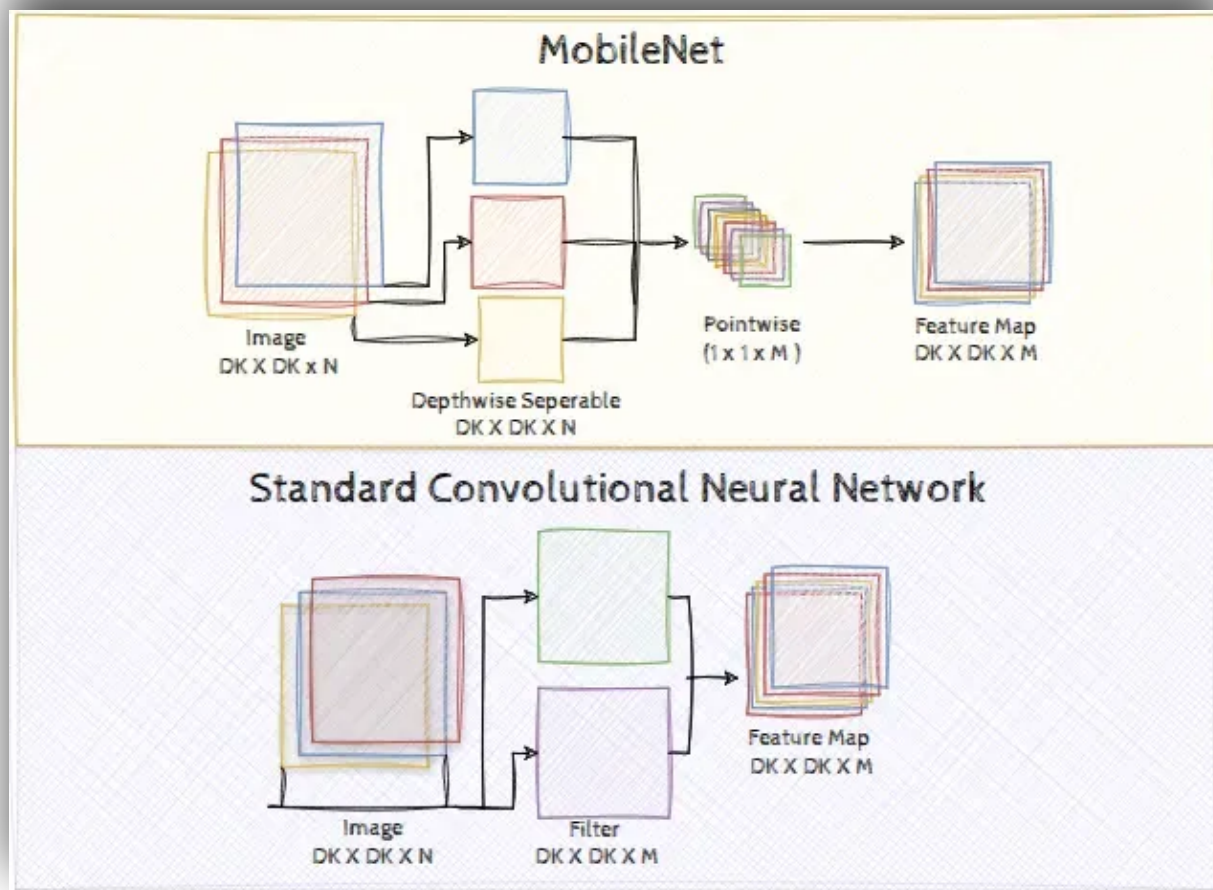
Case Study: Smartphone Vision

Problem: A smartphone application identifies real-time objects through its camera.

Constraints: Must run at 30 FPS on a low-power phone processor without draining the battery.

Solution: Use MobileNet as the backbone.

Why? A heavy ResNet-101 or EfficientNet-B7 would cause massive lag and quickly overheat the phone.



You have 2,000 labeled images and need to build an image classifier. Would you train from scratch or start with a pretrained model? Why?

Think — Pair — Share

Quick Concept Check

- ? What does it mean to train a model "from scratch"?
- ? Explain the concept of Transfer Learning in your own words.
- ? What is a pretrained model?
- ? What is the difference between a backbone and a head?
- ? In what physical scenario would you strictly choose MobileNet over ResNet-152?

One-Minute Recap



Transfer Learning

Reuse learned knowledge rather than training from scratch.



Backbone & Head

Pretrained backbone extracts features; new head classifies.



Architectures

ResNet (Skips), EfficientNet (Scale), MobileNet (Edge).

"

If you had only 1,000 labeled images to build a system tomorrow, would you start from random weights?

"

Next Lecture: Feature Extraction vs Fine-Tuning

Image Sources



<https://kims.org.uk/wp-content/uploads/2024/11/x-ray-scan.png>

Source: kims.org.uk



https://upload.wikimedia.org/wikipedia/en/f/fc/Cats_%26_Dogs_film.jpg?utm_source=en.wikipedia.org&utm_campaign=index&utm_content=original

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